

Assignment 11: Image Classification Using Random Forest

Introduction:

The objective of this project was to develop an image classification model using the Random Forest algorithm. The provided dataset contains images from five different categories: Dalmatian, Dollar Bill, Pizza, Soccer Ball, and Sunflower. The project followed a complete machine learning workflow, including image preprocessing, model training, evaluation, feature importance visualization, prediction on new images, and comparison with an SVM classifier.

Dataset Description

The dataset was provided with the assignment as a compressed ZIP file containing five image categories. Before training the model, all images were extracted and organized by class. The dataset was explored by counting the number of images in each category and visualizing their distribution using a bar chart.

Image preprocessing included:

- Loading the image files.
- Resizing all images to 64×64 pixels.
- Converting images to RGB format.
- Normalizing pixel values by dividing them by 255.
- Flattening each image into a one-dimensional feature vector.
- Splitting the dataset into 80% training data and 20% testing data.



Methodology

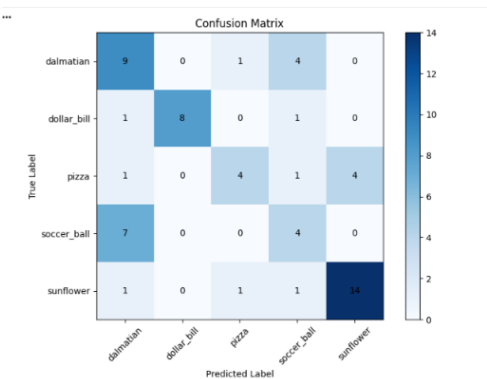
A Random Forest classifier was selected as the primary machine learning model. Instead of using the default parameters, GridSearchCV was applied to identify the best hyperparameter combination. The optimized model was then trained using the training dataset.

The model was evaluated using Accuracy, Precision, Recall, F1-score, Classification Report, and Confusion Matrix. Feature importance values were extracted from the trained model and the twenty most important features were visualized. A prediction function was also implemented to classify new images using the same preprocessing pipeline.

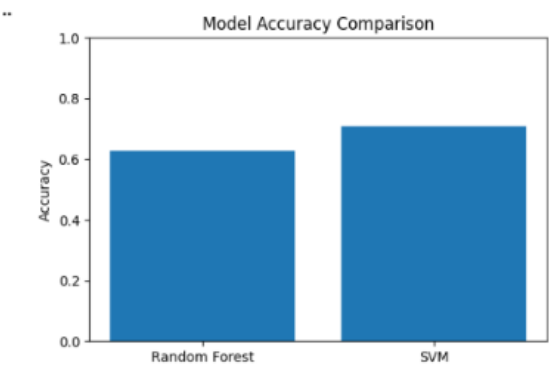
As a bonus task, an SVM classifier was trained using the same training and testing datasets. The performance of both models was compared using classification accuracy

Results

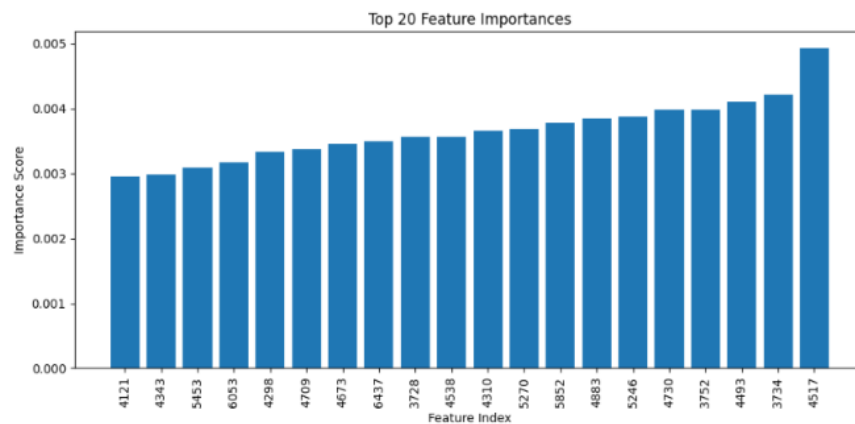
The Random Forest model achieved an overall accuracy of **62.90%** on the testing dataset. The confusion matrix and classification report showed that the model was able to classify most image categories correctly, although some confusion existed between visually similar classes.



The SVM model achieved an accuracy of **70.97%**, outperforming the Random Forest model on this dataset. This comparison demonstrates that different machine learning algorithms can produce different classification performance even when trained on the same data.



The feature importance visualization identified the pixel features that contributed the most to the Random Forest classification process.



Deployment Strategy

The trained model can be deployed as a web application or a mobile application where users upload an image and receive the predicted category. Before prediction, the uploaded image must undergo the same preprocessing steps used during training. To improve performance over time, the model should be periodically retrained using additional labeled image data.

Conclusion

This project successfully demonstrated the complete workflow of an image classification system using Random Forest. Image preprocessing, hyperparameter tuning, model evaluation, feature importance visualization, and prediction on new images were successfully implemented. Finally, the Random Forest model was compared with an SVM classifier, and the results showed that the SVM model achieved higher classification accuracy on this dataset.